

VARADIA DE MURES, Romania

WMO: 152040

Lat: 46.0192N

Lon: 22.1508E

Elev: 515

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	15.8	5.5	7.4	14.7	11.3	10.0	18.7	13.5	37.7	9.8	34.0	1.1	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	26.0	91.4	70.8	88.0	70.3	84.8	69.0	73.3	85.3	71.8	84.2	70.4	82.3	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.7	111.3	78.9	68.1	105.3	76.0	66.7	100.2	74.3	37.2	85.2	35.8	84.3	34.6	82.6	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	10.0	8.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	30.2	36.1	42.6	51.5	60.4	67.1	68.9	70.0	60.5	51.6	43.3	34.4
	DBStd	15.02	7.99	7.01	6.82	8.00	5.48	5.41	5.03	5.25	6.77	6.79	7.34	6.59
	HDD50	2106	613	389	243	79	3	0	0	3	65	225	486	
	HDD65	5408	1078	809	694	405	160	40	21	17	167	415	652	950
	CDD50	2637	0	0	14	125	325	512	587	620	316	115	23	0
	CDD65	466	0	0	0	0	17	103	143	172	31	0	0	0
	CDH74	5806	0	0	2	109	343	1154	1669	2000	459	69	1	0
	CDH80	2074	0	0	0	6	54	362	637	875	137	3	0	0
(14) Wind	WSAvg	2.2	1.5	1.9	2.7	3.0	2.6	2.5	2.6	2.4	2.0	1.8	1.6	1.6
(15) Precipitation	PrecAvg	28.6	1.9	1.6	1.6	2.3	3.0	3.3	3.3	2.1	2.7	2.5	2.1	2.4
	PrecMax	39.1	3.6	4.8	3.5	3.6	6.7	6.4	7.4	4.1	6.5	5.3	4.6	5.1
	PrecMin	19.6	0.6	0.2	0.4	0.7	1.3	1.2	1.0	0.6	0.3	0.1	0.0	0.2
	PrecStd	5.3	0.9	1.1	1.0	0.8	1.2	1.3	1.8	1.1	1.3	1.3	1.0	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	62.0	72.1	80.6	85.1	91.1	95.2	96.0	89.3	79.9	69.1	55.7
		MCWB	47.3	50.5	54.8	61.3	66.3	72.0	70.4	70.7	66.9	61.0	55.9	49.3
	2%	DB	47.0	57.0	66.5	77.4	81.4	87.8	90.5	92.3	83.5	75.5	64.2	49.5
		MCWB	42.9	48.0	51.9	59.7	65.8	72.0	70.8	70.0	65.9	60.3	54.1	45.7
	5%	DB	42.8	52.2	60.8	73.2	78.0	84.4	86.7	88.3	79.6	70.9	59.6	45.4
		MCWB	40.5	44.9	49.0	58.3	63.8	69.6	70.1	70.0	65.2	58.6	52.1	43.0
	10%	DB	39.9	47.6	56.1	68.5	74.6	81.3	82.9	84.7	74.1	64.9	54.8	42.5
		MCWB	38.1	42.8	46.6	55.6	62.5	68.8	68.7	69.1	63.1	56.4	50.1	40.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.1	52.4	56.6	64.4	69.9	76.0	74.4	75.3	70.0	63.9	57.4	50.9
		MCDB	52.1	59.9	70.9	77.3	80.0	87.1	85.2	87.6	85.3	74.0	65.9	54.6
	2%	WB	43.8	48.9	53.5	61.5	67.6	73.3	72.7	72.8	67.9	61.6	55.4	45.9
		MCDB	46.4	55.3	64.9	74.6	77.9	84.5	84.9	86.0	80.0	72.2	63.2	48.7
	5%	WB	41.0	46.3	50.5	59.3	65.5	71.4	71.4	71.2	66.1	59.7	53.3	43.6
		MCDB	42.6	51.2	59.1	72.1	75.1	82.0	83.7	84.6	76.3	69.0	58.4	45.6
	10%	WB	38.2	43.2	48.2	56.9	63.5	69.5	69.9	69.8	64.3	57.3	50.8	41.0
		MCDB	39.6	46.8	54.8	67.2	72.5	79.5	81.1	82.9	73.1	64.6	54.8	42.3
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	18.3	22.3	24.2	23.3	23.1	25.1	26.0	22.9	23.6	18.7	11.9
		MCDBR	18.4	27.1	31.4	34.0	30.4	28.5	32.4	33.4	32.1	34.8	26.5	17.0
	5% WB	MCWBR	14.3	18.0	17.6	17.5	16.1	13.4	13.9	13.2	14.4	20.2	16.5	13.0
		MCDWR	16.9	24.6	28.2	30.7	28.2	26.2	27.2	28.8	26.9	31.3	23.8	17.6
(40) Clear-Sky Solar Irradiance	taub		0.332	0.360	0.401	0.457	0.443	0.436	0.452	0.459	0.426	0.385	0.355	0.332
		taud	2.339	2.267	2.205	2.064	2.165	2.227	2.189	2.164	2.224	2.333	2.370	2.368
	Ebn at Noon		239	254	259	254	263	265	258	251	248	242	228	225
		Edh at Noon	24	32	39	49	46	43	45	44	38	30	24	22
(44) All-Sky Solar Radiation	RadAvg		387	645	1017	1407	1692	1876	1850	1683	1189	818	477	310
	RadStd		48	86	141	145	158	162	154	162	130	92	54	42

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	+1.30	N/A	N/A	N/A	+1.71	N/A	N/A	-229	-342	+212	+106

Nomenclature: See separate page